

CO3133 — Deep Learning and Its Applications

Assignment 2 Dataset Proposal

1. Proposed Task

Task: Speech Emotion Recognition (SER)

Task type: Multi-class audio classification

The objective is to classify a speech recording according to the emotion expressed by the speaker.

The project will investigate whether preserving temporal information between convolutional feature extraction and recurrent sequence modeling can improve speech emotion recognition.

The central research hypothesis is:

Removing global temporal average pooling before the recurrent layer and preserving the temporal sequence should allow the GRU to model temporal dependencies more effectively, leading to improved emotion classification performance.

2. Dataset

Dataset: Toronto Emotional Speech Set (TESS)

Source: University of Toronto — Scholaris Repository

Dataset URL:

<https://utoronto.scholaris.ca/collections/036db644-9790-4ed0-90cc-be1dfb8a4b66/search>

License: CC BY-NC 4.0 (Creative Commons Attribution-NonCommercial 4.0 International). The dataset may be used and shared for research and educational purposes with proper attribution, but commercial use is not permitted

Year: 2010

The dataset consists of acted speech recordings labeled according to emotional category.

The final proposal will record the exact dataset metadata, collection identifier, version information if provided by the repository, license/access conditions, and download date.

Before training, the dataset metadata and directory structure will be inspected to determine the exact number of recordings, emotion categories, speakers, and available speaker-level information.

3. Task Input and Output

Input: A speech audio waveform.

The audio will be standardized to a consistent sampling rate and channel configuration before being passed to the models.

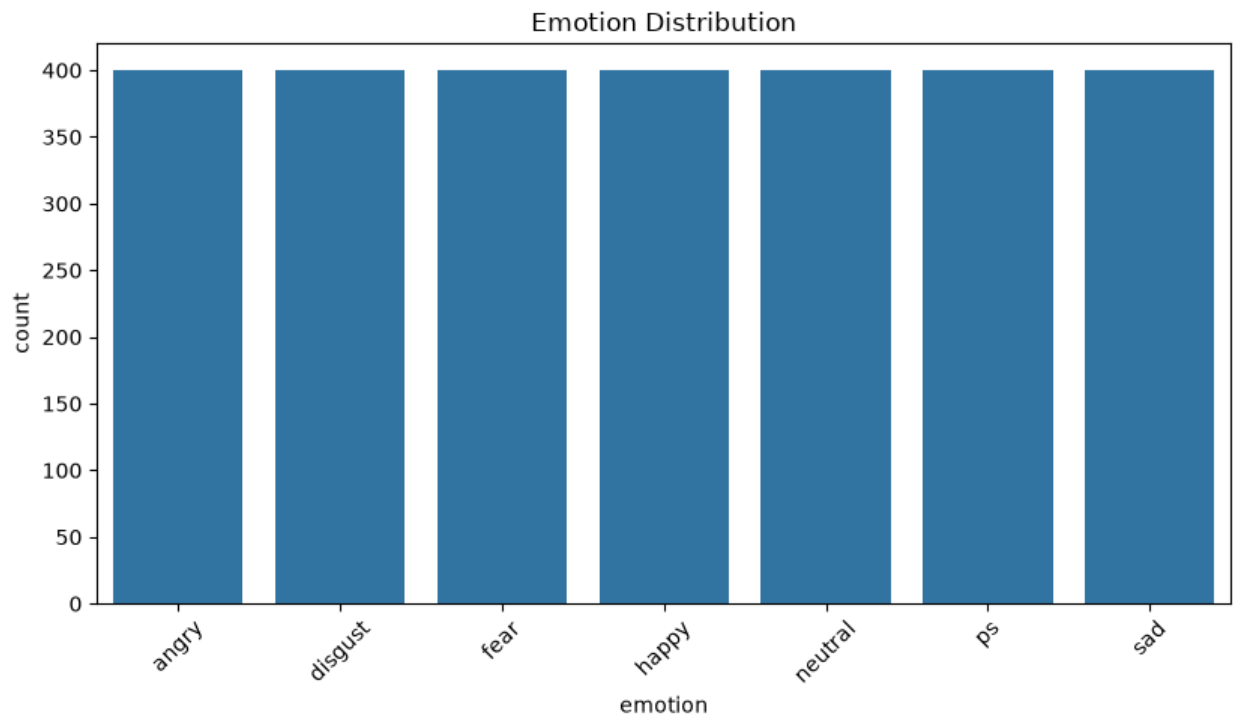
Two input representations will be investigated:

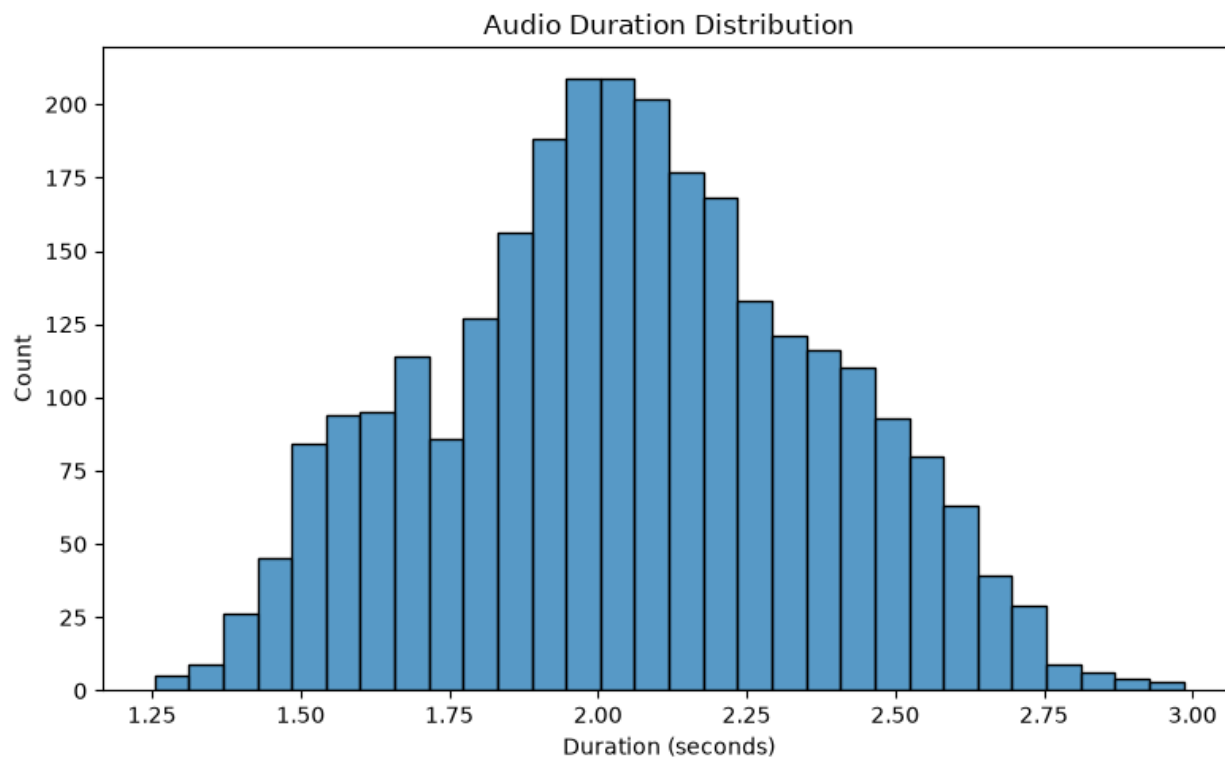
1. **Log-mel spectrogram representation** for the baseline model.
2. **Raw waveform representation** for the CNN-n-GRU reference and modified models.

Output: A probability distribution over the emotion classes represented in the dataset.

The predicted emotion will be the class with the highest predicted probability.

4. Preliminary Data Analysis





The TESS dataset contains 2,800 audio recordings with no corrupted or unreadable files detected. The dataset is perfectly balanced across the seven emotion classes (angry, disgust, fear, happy, neutral, pleasant surprise, and sad), with 400 samples per class, and is equally distributed between the two speakers (1,400 recordings each). The corpus contains 200 unique words, each appearing consistently across speakers and emotions. Audio durations range from 1.25 s to 2.98 s, with an average duration of 2.06 s, indicating relatively uniform utterance lengths. Nearly all recordings share the same sampling rate (24,414 Hz), with only a single outlier file recorded at 96,000 Hz. Overall, the dataset is clean, well-balanced, and suitable for supervised speech emotion recognition experiments.

The following exploratory analysis will be performed before model training:

1. Determine the total number of audio recordings.
2. Determine the number of emotion classes.
3. Analyze the number of samples per emotion.
4. Analyze speaker distribution across emotion classes.
5. Analyze audio duration distribution.
6. Analyze sampling rates and audio channels.

7. Inspect representative waveforms and spectrograms.
8. Analyze the class balance.
9. Identify potential duplicate or highly similar recordings.
10. Determine whether speaker identity can be used to construct a leakage-resistant split.

Particular attention will be paid to **speaker distribution**, because speech emotion recognition models can learn speaker-specific characteristics rather than emotion-related characteristics.

5. Data Split and Leakage Prevention

The dataset will be divided into training, validation, and test sets while preserving the emotion-class distribution.

The TESS dataset contains recordings from only two speakers, making a strict speaker-level train/validation/test split impractical. Therefore, stratified splitting will be performed at the **recording level** while preserving class distributions across all partitions. Since recordings from the same speakers may appear in multiple splits, results will be interpreted with the understanding that speaker-specific characteristics may contribute to model performance.

The proposed split is:

- **70% training**
- **15% validation**
- **15% testing**

The exact split strategy will be finalized after inspecting the dataset's speaker metadata.

A fixed random seed will be used to make the split reproducible.

The test set will remain completely isolated from model selection and hyperparameter tuning and will only be used for final evaluation.

6. Evaluation Metrics

The primary evaluation metrics will be:

- **Accuracy**

- **Macro-F1 score**

Accuracy provides an overall measure of classification correctness, while Macro-F1 gives equal importance to each emotion class.

Additional evaluation will include:

- Per-class precision
- Per-class recall
- Per-class F1-score
- Confusion matrix

These metrics will allow the project to determine which emotions are most difficult to distinguish.

7. Baseline Model

A conventional spectrogram-based deep-learning pipeline will be implemented as the simple baseline:

Raw audio waveform → Log-mel spectrogram → 2D CNN → GRU/LSTM → classifier

The spectrogram converts the one-dimensional audio waveform into a time-frequency representation. The 2D CNN will extract local time-frequency features, while the recurrent layer will model temporal dependencies between extracted feature representations.

The baseline will be trained from scratch on the training portion of the dataset.

Its purpose is to establish a reference performance for a conventional speech-emotion-recognition pipeline.

8. Modern / Reference Model

The project will use the **CNN-n-GRU architecture** proposed by Nfissi et al. as the modern/reference architecture.

Repository:

<https://github.com/alaaNfissi/CNN-n-GRU-for-speech-emotion-recognition>

The architecture processes raw speech waveforms directly using convolutional layers followed by recurrent layers. The authors evaluate CNN-n-GRU variants with different numbers of convolutional and recurrent layers, including CNN-3-GRU, CNN-5-GRU, CNN-11-GRU, and CNN-18-GRU.

The repository reports experiments on TESS, IEMOCAP, and RAVDESS and states that the models were trained from scratch rather than initialized from pretrained weights.

For this project, one CNN-n-GRU configuration will be selected as the reference architecture based on preliminary experiments and computational feasibility.

9. Proposed Architectural Modification

The primary modification concerns the interface between the convolutional feature extractor and the recurrent layer.

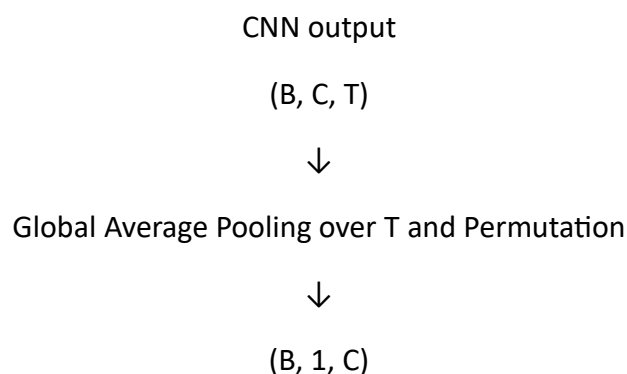
In the reference architecture, convolutional layers produce a feature representation with temporal structure. The architecture then performs global average pooling over the remaining temporal dimension before passing the representation to a fully connected layer and subsequently to the GRU.

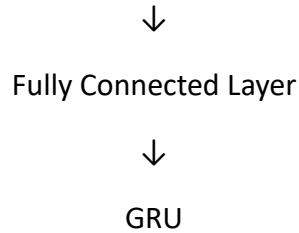
This operation compresses the temporal dimension into a single feature vector. Consequently, the recurrent layer receives only a single effective timestep and therefore cannot fully exploit its intended sequential modeling capability.

The proposed modification removes this global temporal average pooling operation.

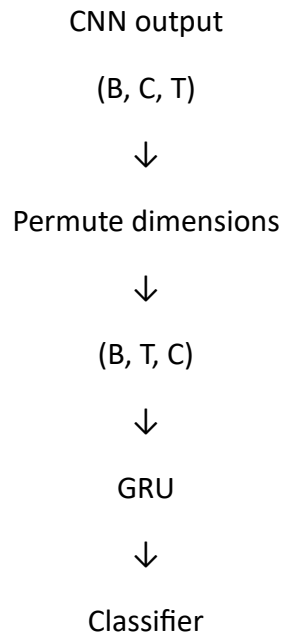
Instead, the temporal dimension will be preserved:

Reference:





Proposed:



The resulting GRU will therefore receive the CNN representation as an actual sequence of feature vectors.

The hypothesis is that this modification will allow the recurrent component to capture temporal relationships between acoustic patterns occurring at different points in the speech signal.

10. Main Research Question

The primary research question is:

Does preserving the temporal dimension between the CNN feature extractor and GRU improve speech emotion recognition compared with the original CNN-n-GRU architecture?

The corresponding hypothesis is:

H1: A CNN-GRU architecture that preserves the temporal sequence of convolutional features will achieve higher emotion-classification performance than the original architecture that applies global temporal average pooling before the GRU.

The null hypothesis is:

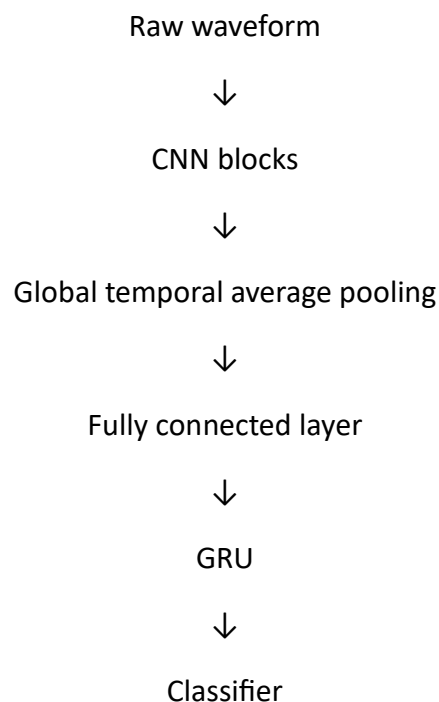
H0: Removing the temporal global-average-pooling operation does not produce a meaningful improvement in speech emotion recognition performance.

11. Controlled Experiment

The primary controlled experiment will compare two architectures.

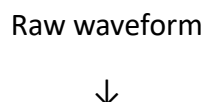
Model A — Original CNN-n-GRU

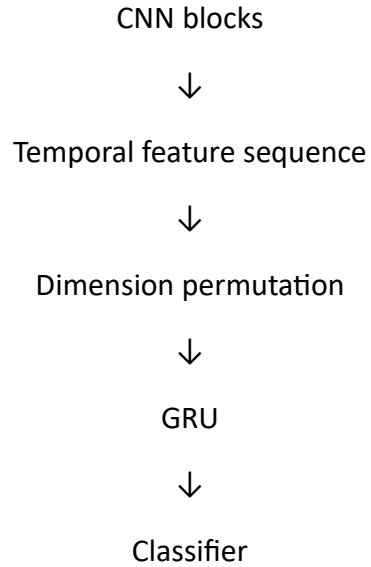
The original architecture will be reproduced as closely as possible.



Model B — Modified CNN-n-GRU

The proposed modification will remove the temporal global average pooling operation.





The following factors will remain fixed between the two models:

- Dataset
- Train/validation/test split
- Audio preprocessing
- Sampling rate
- CNN architecture
- GRU architecture
- Hidden dimension
- Number of recurrent layers
- Optimizer
- Learning rate
- Batch size
- Number of training epochs
- Data augmentation
- Random seed
- Checkpoint-selection criterion

The primary changed factor will be the **presence or absence of temporal global average pooling before the recurrent layer.**

12. Fine-Tuning / Training Protocol

Because the selected CNN-n-GRU reference implementation is trained from scratch rather than initialized from pretrained weights, the project will treat it as a modern/reference architecture rather than a pretrained backbone.

The training protocol will therefore be explicitly documented and controlled.

The following parameters will be reported:

- Weight initialization
- Optimizer
- Initial learning rate
- Learning-rate schedule
- Batch size
- Number of epochs
- Regularization
- Dropout
- Data augmentation
- Model-selection criterion
- Random seed
- Hardware configuration

The modified architecture will be trained under the same protocol as the reference architecture to ensure that performance differences can be attributed primarily to the architectural modification.

13. Quantitative Evaluation

The reference and modified models will be compared using:

Metric	Purpose
Accuracy	Overall classification performance
Macro-F1	Balanced performance across emotion classes

Per-class F1	Identify class-specific improvements
Parameter count	Measure model complexity
Training time	Measure computational cost
Inference time	Measure deployment cost

The final test-set results will be reported only after model architecture and training configuration have been finalized using the training and validation sets.

14. Qualitative Evaluation and Error Analysis

A confusion matrix will be generated for each major model.

Misclassified recordings will be analyzed to identify common error patterns, including:

- Confusion between acoustically similar emotions.
- Emotion classes with consistently low recall.
- Recordings with unusual speech characteristics.
- Errors potentially caused by speaker characteristics.
- Errors associated with temporal patterns that may be lost by the reference architecture.

Where appropriate, representative waveform and spectrogram visualizations will be used to compare correctly and incorrectly classified examples.

Particular attention will be given to whether the modified model reduces errors involving emotions whose distinguishing characteristics occur at different points in the speech sequence.

15. Compute Estimate

The experiments will require GPU acceleration because the project involves raw-waveform convolutional processing and recurrent sequence modeling.

The expected requirements are:

- 1 GPU
- Approximately 8–16 GB GPU memory preferred
- PyTorch-based implementation

- Multiple training runs for baseline, reference, and modified architectures

The exact hardware, software versions, training duration, parameter counts, and inference cost will be recorded during experimentation.

A smaller subset of the training data may be used during initial debugging and architecture validation, but final experiments will use the complete approved dataset where computationally feasible.

16. Expected Contribution

The project aims to determine whether the temporal information discarded by global average pooling is important for speech emotion recognition.

Rather than proposing an entirely new neural architecture, the project focuses on a targeted architectural intervention:

Preserve the temporal sequence produced by the CNN and allow the GRU to operate on the sequence directly.

If the proposed modification produces a consistent improvement, the results will provide evidence that the original pooling operation limits the effectiveness of the recurrent component.

If the modification does not improve performance, the experiment will investigate whether the temporal information removed by pooling is actually useful for this particular task or whether other factors dominate performance.

The final conclusion will therefore be based on controlled quantitative experiments, qualitative error analysis, and computational-cost comparisons.